

Relationship Intelligence Workbench

Back-End Requirements Document for the SIX Wealth Advisory Challenge

Version 1.0 | Technical, Legal, Financial, and Product Requirements

Purpose. This document converts the product concept into implementable back-end requirements. It is designed for a hackathon MVP, but it is written with a production-grade wealth-management mindset: explainability, suitability, privacy, auditability, and human control are treated as core architecture, not add-ons.

Legal note. This is a product and engineering requirements document, not legal advice. Any production deployment would require review by legal, compliance, risk, data protection, information security, and investment governance teams.

0. Source Basis and Non-Negotiable Product Direction

The challenge asks teams to build a next-generation advisor dashboard where a relationship manager can understand client investment identity, monitor portfolio events, news, and CIO signals, and act on the right insight at the right time. The strategy stays unchanged, while personalization happens at the asset level within Defensive, Balanced, or Growth mandates. The AI must equip the RM, not advise the client directly.

Source	Requirement implication
SIX challenge brief	Client-first dashboard, Client DNA from CRM notes, portfolio/news/CIO monitoring, same-sector asset swaps within the existing mandate, tailored RM messages, human in the loop.
SIX values	Design must express customer focus, collaboration, ownership, and trust. In back-end terms: evidence, traceability, approval workflows, and transparent controls.
Swiss and EU data protection principles	Only process necessary data for explicit purposes. Collection, storage, retrieval, and profiling are already regulated processing. Avoid personal social-media scraping.
FinSA and MiFID suitability logic	Investment actions must be suitability-aware: financial situation, objectives, knowledge and experience, risk tolerance, and ability to bear losses must be respected.
FINMA AI guidance	AI risk controls must cover robustness, correctness, bias, explainability, cyber risk, third-party dependency, and accountability.

1. Product Vision and Back-End Mission

Back-end mission: transform relationship-manager notes into a living Client DNA, connect that DNA to portfolio, market, news, and CIO signals, then generate proactive, explainable, suitability-aware actions for RM approval.

Back-end principle	Meaning for implementation
No citation, no action	Every alert, recommendation, and client story must link to source evidence: RM note, holding, news event, CIO signal, or strategy rule.
Human approval by design	AI outputs remain draft-only until the RM approves. The system must not execute orders or contact clients autonomously.
Suitability before personalization	A client-preference match is not enough. The action must respect mandate, risk, liquidity, concentration, knowledge and experience, and CIO constraints.
Progressive DNA	The system may infer relationship signals, but must label explicit, repeated, inferred, and missing fields with confidence levels.
Privacy-minimal personalization	Use CRM notes, bank-held profile data, portfolio data, CIO lists, and approved news/market feeds. Exclude personal social-media scraping from MVP scope.
Relationship value without manipulation	The system may help expand share of wallet over the client lifetime by improving relevance and trust, but must not optimize for sales over suitability.

2. Scope

In scope for MVP	Out of scope for MVP
Ingest CRM notes, portfolios, CIO recommendation list, SIX market/reference data, and news events.	Direct client advice, autonomous order placement, or sending messages without RM approval.
Build Client Memory Card, Client DNA, communication style tags, DNA gaps, and source-backed evidence.	Scraping LinkedIn, Instagram, X, Facebook, TikTok, or other personal social media.
Generate anticipation alerts, predicted client questions, same-sector asset swap candidates, and client-specific message drafts.	Replacing the bank suitability process or legal compliance function.
Provide audit logs, approval states, rationale, and documentation exports.	Full production-grade portfolio optimization engine or tax engine.
Support hackathon personas and demo scenarios using provided workbooks.	Real client data ingestion unless explicitly approved by the organizers and relevant data owners.

3. Target Back-End Architecture

Recommended MVP architecture: TypeScript/Express or Python/FastAPI back end, relational database, lightweight vector store, model-orchestrated agents, and deterministic scoring services. The model should generate structured JSON, not free-form logic, wherever possible.

Layer	Components	Responsibilities
Data ingestion layer	CRM importer, portfolio importer, CIO list importer, news connector, SIX MCP/Web API connector	Normalize source data, attach source IDs, classify data sensitivity, and store immutable raw references.
Client intelligence layer	Client DNA service, Memory Card service, communication-style classifier, DNA Gap Finder	Convert notes into structured, confidence-aware relationship intelligence.
Signal layer	News/CIO/portfolio event service, event severity classifier, portfolio materiality engine	Identify relevant events and calculate whether they matter for a specific client.
Action layer	Alert orchestrator, swap candidate engine, suitability guardrails, Client Story Builder	Produce next-best-actions and messages that answer: Why now, why this client, why this action, how to say it.
Control layer	Evidence Ledger, approval workflow, audit log, data-permission registry, policy engine	Ensure explainability, legal traceability, human control, and compliance documentation.
Presentation API layer	REST endpoints for dashboard, alerts, client page, messages, approvals, and documents	Serve UI-ready payloads with citations, confidence scores, and action states.

4. Data Requirements

All data objects must include source metadata, ingest timestamp, allowed-use status, and sensitivity classification. The system should distinguish bank-held data from external market/news data and should never silently mix source categories.

Data source	Allowed use	Required metadata
CRM notes	Extract Client DNA, relationship memory, communication style, preferences, sensitivities, DNA gaps.	client_id, rm_id, note_id, date, author, source_system, source_snippet, consent_or_bank_basis, sensitivity_label.
Portfolio holdings	Calculate exposure, materiality, mandate drift, concentration, holding affected by news.	client_id, portfolio_id, ISIN/Valor/MIC/ticker, asset class, sector, country, currency, quantity, current_value_chf, target_value_chf, weight.

Portfolio strategy	Check mandate, asset-class ranges, target allocation, drift, risk category.	mandate_type, target weights, allowed ranges, drift threshold, strategy date, source workbook tab.
CIO recommendation list	Constrain eligible swap universe and ratings.	instrument_id, CIO rating, sector, asset class, mandate suitability, date, rationale if provided.
SIX data	Reference data, prices, fundamentals, macro data, issuer or instrument attributes.	instrument identifier, timestamp, data vendor, endpoint/tool, currency, price date.
News/events	Trigger event severity, relevance, sentiment, controversy type, company/sector linkage.	event_id, source, publish time, entities, sentiment, reliability, URL or provider reference, raw summary.
Client consent/permission	Control which data categories can be used and whether outreach is permitted.	client_id, permission_type, status, date, scope, expiry, reviewer.

5. Data Model

Entity	Key fields	Notes
Client	client_id, name, segment, mandate_type, rm_id, AUM, risk_profile, status	For demo, client names may be personas. In production, pseudonymize where possible.
RMNote	note_id, client_id, date, text, author, source_system, sensitivity_label	Raw notes should be access-controlled and never exposed to the LLM without minimization.
DNA	attribute_id, client_id, category, value, confidence, signal_type, evidence_ids, last_updated	Categories include values, goals, business context, family context, preferences, sensitivities, communication style.
PortfolioHolding	holding_id, client_id, instrument_id, quantity, current_value_chf, target_value_chf, weight, drift_pp	Current and target values allow mandate drift checks.
Instrument	instrument_id, ISIN, Valor, MIC, ticker, issuer, sector, asset_class, currency, risk_score, tags	Instrument-level metadata powers same-sector and preference matching.
NewsEvent	event_id, entities, sector, severity, sentiment, topic_tags, source, published_at	Events should be deduplicated and linked to holdings and preference tags.
CIOSignal	signal_id, instrument_id, rating, effective_date, mandate_allowed, rationale	Candidate universe should be constrained to approved instruments.
Alert	alert_id, client_id, event_id, priority_score, status, why_now, why_client, evidence_ids	Alert is not a recommendation until suitability and action checks pass.
Recommendation	recommendation_id, alert_id, current_holding_id, candidate_id, action_type, suitability_status, fit_score, rationale	Action types: HOLD_WITH_EXPLANATION, WATCH, SWAP, REBALANCE, CALL_CLIENT.
MessageDraft	message_id, recommendation_id, style_tag, subject, opening, situation, recommendation, next_step, approval_status	Always draft-only before RM approval.
ApprovalEvent	approval_id, object_id, rm_id, action, timestamp, edits_summary, final_text_hash	Required for audit and accountability.
EvidenceItem	evidence_id, source_type, source_id, snippet, timestamp, reliability_score, allowed_use	Every AI output must cite evidence items.
AuditLog	audit_id, actor_id, action, object_type, object_id, before_hash, after_hash, timestamp	Immutable append-only log.

6. Functional Requirements

FR-001. CRM Note Ingestion

- Requirement: The system shall ingest CRM notes per client and preserve note-level provenance, including date, author, source tab/system, and note ID.
- Acceptance criteria: Given a client, the API returns all derived DNA signals with evidence links back to specific notes or note snippets.

FR-002. Client DNA Extraction

- Requirement: The system shall extract structured Client DNA attributes from RM notes across categories: values, goals, business context, family context, preferences, sensitivities, communication style, and relationship memory.
- Acceptance criteria: At least ten meaningful attributes are extracted across the four demo personas, with confidence scores and source references.

FR-003. DNA Signal Typing

- Requirement: The system shall classify each DNA attribute as explicit preference, repeated pattern, inferred hypothesis, or missing field.
- Acceptance criteria: Low-confidence inferred signals are not used as hard recommendation constraints unless confirmed by the RM.

FR-004. Client Memory Card API

- Requirement: The system shall expose a compact client summary for the UI with what matters, personal context, communication style, previous objections, open promises, and sensitive topics.
- Acceptance criteria: GET /api/clients/{id}/memory-card returns a UI-ready object in under 500 ms from cached data.

FR-005. DNA Gap Finder

- Requirement: The system shall detect missing or weak client-knowledge areas and generate RM-friendly questions to improve future notes.
- Acceptance criteria: At least three useful gap questions are produced for a client whose CRM notes lack sustainability, liquidity, sector exclusion, or next-generation planning information.

FR-006. Portfolio Materiality Engine

- Requirement: The system shall calculate holding weight, target drift, asset-class exposure, sector exposure, currency exposure, and concentration flags.
- Acceptance criteria: For each current holding, the system returns current value, weight, target value, drift in percentage points, and materiality band.

FR-007. News and CIO Signal Linking

- Requirement: The system shall link news events and CIO signals to holdings by instrument, issuer, sector, geography, and topic tags.
- Acceptance criteria: A news event related to a held issuer produces at least one linked holding and one candidate alert.

FR-008. Anticipation Alerts

- Requirement: The system shall rank alerts by predicted client concern, not only market severity.
- Acceptance criteria: An alert explains why the RM should reach out before the client does and includes a predicted client question.

FR-009. Four-Question Action Card

- Requirement: Every action card shall answer: Why now, why this client, why this action, and how to say it.
- Acceptance criteria: No alert can be displayed in high priority unless all four fields are populated or explicitly marked unavailable.

FR-010. Same-Sector Asset Swap Engine

- Requirement: For a conflicted holding, the system shall propose eligible same-sector alternatives from the CIO-approved universe while preserving mandate and suitability constraints.
- Acceptance criteria: The recommendation payload shows current holding, proposed candidate, sector match, CIO rating, risk delta, mandate impact, and evidence.

FR-011. Suitability Guardrails

- Requirement: The system shall run hard-stop and warning checks before showing any recommendation as actionable.
- Acceptance criteria: If a candidate breaches mandate, risk, liquidity, knowledge/experience, or CIO constraints, action_status is BLOCKED or REVIEW_REQUIRED.

FR-012. Client Story Builder

- Requirement: The system shall generate a structured message draft with opening, situation, recommendation, client-specific rationale, evidence, and next step.
- Acceptance criteria: Drafts adapt to at least five communication styles: numbers-first, narrative-first, concise, risk-sensitive, and values-driven.

FR-013. Advisor Action Pack

- Requirement: The system shall generate meeting preparation, presentation outline, follow-up draft, and internal documentation from the same evidence base.
- Acceptance criteria: All generated materials cite the same alert_id and recommendation_id and can be regenerated after RM edits.

FR-014. Human Approval Workflow

- Requirement: The system shall require RM approval before a message or recommendation is marked ready for client contact.
- Acceptance criteria: A message cannot transition from DRAFT to APPROVED without rm_id, timestamp, final text hash, and approval event.

FR-015. Audit and Documentation

- Requirement: The system shall document the client needs, recommendation basis, sources, suitability checks, RM approval, and final output.
- Acceptance criteria: Audit log can reconstruct why a recommendation was made, edited, approved, rejected, or blocked.

7. Scoring System and Financial Parameters

All thresholds below are configurable demo defaults, not universal investment advice. Store them in a versioned policy configuration file so the team can adjust the demo without changing code.

Parameter	Refined threshold or rule	Backend implementation and control logic
Threshold governance	All thresholds are configurable, versioned, and owned by policy/risk/compliance	Store each rule with 'rule_id', 'version', 'owner', 'effective_date', and 'rationale'. These are demo defaults, not universal investment rules.
DNA confidence	High ≥ 0.80 ; Medium 0.55 to 0.79; Low 0.35 to 0.54; Ignore < 0.35	High may drive alerts, recommendations, and message personalization. Medium may influence alert ranking and communication style, but cannot be the sole reason for an asset swap unless the RM confirms it. Low appears only as a hypothesis requiring RM confirmation. Ignore values are not shown to the RM.
Note staleness decay	Reduce DNA confidence after 12 months unless reconfirmed	Recent notes receive higher weight. Older preferences decay unless repeated or confirmed by the RM. This prevents outdated client preferences from driving current recommendations.
Evidence minimum	No action card unless at least one source snippet and one portfolio/security data point exist	Every alert must show evidence from RM notes, portfolio holdings, news, CIO signal, or approved market data. If evidence is missing, the item becomes a DNA Gap or RM review item instead of a recommendation.
Alert priority	Critical ≥ 85 ; High 70 to 84; Watch 50 to 69; Low Priority / Archived < 50	Critical and High appear on the all-clients dashboard. Watch appears on the client page. Low Priority / Archived does not interrupt the RM but remains searchable and auditable. Compliance or suitability issues must never be suppressed only because commercial priority is low.
Alert priority formula	Weighted score based on materiality, DNA relevance, news severity, CIO signal, time sensitivity, and predicted client concern	Suggested formula: 'Alert Priority = portfolio materiality + DNA relevance + news severity + CIO signal strength + time sensitivity + predicted client concern'. Weights should be configurable.
Portfolio materiality by holding weight	Watch 1% to 2%; Actionable 2% to 5%; High 5% to 10%; Concentrated $> 10\%$	Higher materiality increases alert priority and documentation requirements. Use both portfolio percentage and absolute currency exposure, because a small percentage may still represent a large CHF/EUR/USD amount for HNWI/UHNWI clients.
Absolute exposure materiality	Configurable by bank, for example: Watch $\geq 100k$; Actionable $\geq 500k$; High $\geq 1m$; Critical $\geq 5m$	The system should combine percentage weight and absolute value. Example: a 1.5% position may become High Priority if the absolute exposure is large and strongly linked to Client DNA.
Look-through exposure	Use fund/ETF/security look-through where available	If the client has indirect exposure through a fund, ETF, structured product, or mandate sleeve, the system should flag it only when reliable look-through data exists. If unavailable, show "look-through unavailable" rather than assuming no exposure.

Parameter	Refined threshold or rule	Backend implementation and control logic
Mandate drift	Default ± 2.0 percentage points from target allocation	Flag drift breaches. Do not propose swaps that worsen a mandate breach unless the recommendation is explicitly labeled as a rebalancing exception and requires RM approval. Tolerances should be configurable by mandate and asset class.
Single-name / issuer concentration warning	Defensive > 10%; Balanced > 12%; Growth > 15%	Existing positions trigger warning. New or enlarged positions above the threshold require hard review. Check issuer/group exposure, not only security name.
Broader economic concentration	Flag when issuer, sector, country, theme, or client business exposure creates hidden concentration	Example: if the client owns a private luxury business and also has high public luxury-sector exposure, the system should identify the combined economic exposure as a relationship-specific risk.
Sector exposure delta after swap	Preferred ≤ 1.0 pp; Warning > 1.0 pp; Block > 2.0 pp unless rebalancing case	Keeps “same-sector swap” from becoming an unintended sector bet. Define sector taxonomy clearly, for example GICS, ICB, or bank internal taxonomy. Same peer group is preferred over broad same-sector matching.
Risk score delta after swap	Preferred $\leq +0.25$ on 1 to 5 scale; Warning +0.25 to +0.50; Block > +0.50	Risk score must come from bank-approved analytics or approved investment data, not from the LLM. If unavailable, show “risk score unavailable” and require RM review.
Relative volatility or VaR increase	Warning > +2% relative increase; Block > +5% relative increase	Use relative increase, not percentage-point increase. Example: VaR rising from 10.0% to 10.4% is a 4% relative increase. If volatility or VaR data is unavailable, require RM review rather than treating it as safe.
Liquidity horizon mismatch	Hard block if product liquidity or expected holding period conflicts with known cash need within 90 days	Prevents illiquid or unsuitable recommendations when a cash need is known. Connect to Client Memory Card, for example property purchase, tax payment, business liquidity event, or family transfer.
Medium-term liquidity warning	Warning if known or likely cash need exists within 6 to 12 months, depending on product liquidity	Especially relevant for structured products, private markets, alternatives, thinly traded securities, or bonds with limited liquidity. If liquidity needs are missing, DNA Gap Finder asks the RM to confirm.
CIO eligibility	BUY is allowed for new recommendations; approved HOLD only if bank policy permits; SELL / Restricted / Not Covered are blocked	Candidate universe must be constrained by CIO-approved lists. The AI cannot invent investment candidates outside the approved universe.
Restricted list and product eligibility	Block if asset is restricted, sanctioned, banned, unsuitable for client type, or conflicts with client-specific exclusions	Check regulatory restrictions, internal restricted lists, product governance rules, client exclusions, and mandate eligibility before showing a candidate as actionable.
News severity	1 informational; 2 minor; 3 meaningful; 4 major; 5 crisis	Severity combines with DNA relevance, portfolio materiality, source credibility, recency, confirmation status, and entity matching. Rumor or weak-source news should be downgraded or sent to RM review.
News entity matching	Alert only when the news is reliably linked to a held company, issuer, sector, or approved candidate	Prevent false alerts from ambiguous company names, subsidiaries, unrelated tickers, or broad sector news. Store matched entity, matching confidence, source, and timestamp.
Client concern trigger	Concern score ≥ 70	Generates anticipation alert and predicted client question. Suggested formula: ‘Concern Score = DNA relevance + portfolio materiality + news severity + past client behavior + communication sensitivity + upcoming meeting/contact timing’.

Parameter	Refined threshold or rule	Backend implementation and control logic
Predicted client question	Generate when concern score ≥ 70 and evidence minimum is met	Example output: “Client may ask whether we should still hold this company after the governance controversy.” The system should prepare a concise RM answer with evidence and possible next step.
Contact fatigue control	Suppress duplicate outreach within 48 to 72 hours unless alert is Critical	Prevents over-contacting the client. If a similar topic was already discussed recently, show as an update inside the client page rather than a new high-priority alert.
Message style confidence	Use style tag if confidence ≥ 0.60 ; otherwise default to professional concise	Avoids over-personalized tone from weak evidence. Style tags may include Numbers-first, Narrative-first, Very concise, Risk-sensitive, Values-driven, Macro-focused, Family-office style, and Avoids technical jargon.
High-severity message control	For severity 4 or 5 events, use balanced professional template regardless of style tag	Even if a client prefers narrative communication, high-severity issues must include factual trigger, risk note, uncertainty, proposed action, and RM approval.
Sensitive inference blocker	Never infer or use sensitive traits such as religion, politics, health, sexuality, ethnicity, or other protected characteristics	The system may only use investment-relevant, consented, bank-held relationship data. Sensitive inferred traits should be blocked from Client DNA, alerts, and message generation.
Personal social media data	Do not scrape, store, or retrieve personal social media data by default	Client DNA should be built from RM notes, onboarding/suitability data, portfolio holdings, CIO lists, and approved market/news feeds. Optional external enrichment should require explicit client consent and bank approval.
RM override reason	Required whenever an RM proceeds despite a warning or block-level control	Store override reason, RM identity, timestamp, rule breached, evidence shown, and final decision. Blocked actions should only proceed where policy explicitly allows override.
Evidence Ledger	Required for every alert, recommendation, message, and documentation output	Store note snippets, news source, portfolio data, CIO signal, rule checks, model prompt/output metadata, RM edits, approval status, and final client-facing message.
Human approval gate	AI may draft, rank, and explain, but cannot send or execute	RM must review, edit, approve, and own every client interaction. The client remains the final decision-maker.
Data unavailable handling	Missing risk, liquidity, look-through, CIO, or suitability data creates RM Review status	Missing data should never be interpreted as safe. The UI should clearly show unavailable fields and suggest what the RM needs to confirm.
Action eligibility status	Output one of: Actionable, Warning, RM Review, Blocked, DNA Gap	This gives the front end a clean status model. “Actionable” means all hard controls passed. “Warning” means proceed with documentation. “RM Review” means missing or uncertain data. “Blocked” means not eligible. “DNA Gap” means more client information is needed.

7.1 Scoring Formulas

Use deterministic scoring around LLM extraction. The LLM should classify and summarize; the application should calculate priority scores using transparent formulas.

Score	Refined formula	Purpose	Control
Client Concern Score	0.30 DNA relevance + 0.25 portfolio materiality + 0.20 event severity + 0.10 recency + 0.10 relationship timing + 0.05 past reaction pattern	Predicts whether the client is likely to care or reach out.	Creates anticipation alerts, but requires evidence minimum.
Actionability Score	0.25 same-sector fit + 0.25 mandate fit quality + 0.20 risk acceptability + 0.15 CIO rating strength + 0.15 data completeness	Determines whether the system can propose an action.	Calculated only after hard gates pass.
Alert Priority Score	0.60 client concern + 0.25 actionability + 0.15 relationship relevance	Ranks the RM all-clients dashboard.	Gated by suitability status; compliance issues bypass commercial scoring.
Asset Fit Score	0.25 mandate fit quality + 0.20 peer/sector fit + 0.20 CIO rating strength + 0.15 DNA alignment + 0.10 risk improvement/neutrality + 0.10 liquidity/market quality	Ranks eligible swap candidates.	Only eligible assets enter ranking.
Relationship Opportunity Score	0.35 client relevance + 0.25 timing relevance + 0.20 unmet need + 0.10 relationship gap + 0.10 next-best-service potential	Prioritizes RM workload and relationship deepening.	Never overrides suitability or client best interest.

8. Legal, Compliance, and Governance Requirements

The back end must assume that CRM notes, profile attributes, DNA scores, communication styles, and inferred preferences are personal data. Collection, storage, retrieval, and profiling are processing activities. The safest design is privacy-minimal, consent-aware, source-backed, and auditable.

Requirement ID	Requirement	Implementation rule
LG-001	Purpose limitation	Store allowed _use for every data object. Do not reuse CRM notes for unrelated analytics or marketing without a separate approved basis.
LG-002	Data minimization	Only send relevant snippets to the LLM. Prefer structured attributes and evidence IDs over raw full-history notes in prompts.
LG-003	No social-media scraping	Exclude personal LinkedIn, Instagram, Facebook, X, TikTok, and personal websites from ingestion unless explicit client-authorized enrichment is separately approved.
LG-004	Sensitive inference control	Do not infer religion, politics, health, sexuality, ethnicity, or similar sensitive traits from notes. If a sensitive topic is directly client-provided and investment-relevant, mark it restricted and require RM confirmation.
LG-005	Suitability record	For every recommendation, store client objective, risk/tolerance check, knowledge/experience basis, financial situation proxy, mandate impact, and rationale.
LG-006	Human involvement transparency	UI and outputs must make clear that AI drafts, RM reviews, RM recommends, and client decides.
LG-007	Auditability	Every model output must store model name/version, prompt template version, input evidence IDs, output hash, and reviewer action.
LG-008	Third-party and API risk	Store which external providers were used for market data, news, and LLM generation. Support fallback or cached mode if a provider is unavailable.
LG-009	Access control	RM can access only assigned clients. Admin and developer views use pseudonymized demo IDs by default.
LG-010	Right to correction/deletion readiness	Data model must support updating or suppressing incorrect DNA attributes without deleting the immutable audit record.

9. AI, RAG, and Model-Control Requirements

Area	Requirement
Structured outputs	LLM calls for extraction and generation must return JSON matching schemas. Invalid JSON must be rejected and retried with a repair prompt.

Retrieval-augmented generation	Message drafts and rationale must be grounded in selected evidence snippets, not the model memory.
No unsupported claims	If the evidence is missing, the output must say “insufficient evidence” and generate a DNA gap or RM question instead of inventing facts.
Prompt minimization	Send note snippets, not entire client histories, unless the user explicitly refreshes DNA.
Model versioning	Store model_provider, model_name, temperature, prompt_version, and schema_version for every call.
Temperature policy	Extraction and scoring: temperature 0.0 to 0.2. Story drafting: 0.3 to 0.6. Never let creative generation change financial facts.
Cost control	Cache DNA extraction and event linking. Do not reprocess all notes for every page load. Use summaries, embeddings, and incremental refresh.
Safety classifier	Before display, run outputs through a rule checker for prohibited claims, missing evidence, direct-advice language, sensitive inference, and unsupported performance promises.

10. API Requirements

The API should expose clean dashboard-oriented objects. The front end should not assemble logic from raw data; it should receive interpreted, auditable, UI-ready structures.

Endpoint	Method	Purpose
/api/health/integrations	GET	Check database, SIX, news, LLM, and cache availability.
/api/rm/{rm_id}/clients	GET	All-clients dashboard with AUM, mandate, last contact, next meeting, alert count, and priority score.
/api/clients/{client_id}	GET	Client profile header, mandate, risk profile, AUM, and relationship status.
/api/clients/{client_id}/memory-card	GET	Client Memory Card with source-backed relationship summary.
/api/clients/{client_id}/dna	GET	Client DNA attributes, confidence, signal type, evidence, and gaps.
/api/clients/{client_id}/dna/refresh	POST	Incremental or full DNA extraction refresh.
/api/clients/{client_id}/portfolio-fit	GET	Holdings, exposure, drift, concentration, and suitability flags.
/api/alerts	GET	Priority alerts filtered by RM, client, status, or severity.
/api/alerts/{alert_id}	GET	Four-question action card, evidence trail, predicted client question.
/api/alerts/{alert_id}/recommendations	POST	Generate or refresh candidate actions and swap alternatives.
/api/recommendations/{id}/message-draft	POST	Generate communication draft in selected style.
/api/recommendations/{id}/action-pack	POST	Generate meeting prep, presentation outline, follow-up, and documentation.
/api/recommendations/{id}/approve	POST	RM approval, rejection, or request changes.
/api/audit/{object_type}/{object_id}	GET	Evidence and audit trail for any alert, recommendation, or message.

11. Core JSON Contracts

The following shapes should be implemented first because they directly support the demo screens.

AlertActionCard example fields:

```
{
  "alert_id": "A-001",
  "client_id": "raeber",
  "priority_score": 88,
  "status": "HIGH_PRIORITY",
  "why_now": {"summary": "CIO suggests rebalancing toward US AI stocks", "evidence_ids": ["cio-14"]},
  "why_this_client": {"summary": "Notes show aversion to US tech and defensive value orientation", "evidence_ids": ["note-33", "note-41"]},
```



```

"why_this_action": {"summary": "Alternative preserves defensive mandate and avoids unintended US tech tilt", "recommendation_id": "R-001"},
"how_to_say_it": {"style": "concise risk-sensitive", "message_id": "M-001"},
"predicted_client_question": "Are we being pushed into US technology despite our stated preference?",
"evidence_trail_url": "/api/audit/alert/A-001"
}

```

Recommendation example fields:

```

{
  "recommendation_id": "R-001",
  "action_type": "SWAP_OR_HOLD_WITH_EXPLANATION",
  "current_holding": {"instrument_id": "...", "weight": 0.042, "sector": "Industrials"},
  "candidate": {"instrument_id": "...", "cio_rating": "BUY", "sector_match": true},
  "suitability_status": "PASS_WITH_WARNINGS",
  "guardrails": [{"rule": "same_mandate", "status": "PASS"}, {"rule": "sector_delta", "status": "PASS"}],
  "asset_fit_score": 82,
  "client_dna_alignment": [{"attribute": "defensive_value", "confidence": 0.86}],
  "rm_required_action": "Review and approve before client contact"
}

```

12. Security, Privacy, and Access Control Requirements

Control	MVP requirement	Production direction
Authentication	Simple login or demo RM selector.	SSO, MFA, session management, device controls.
Authorization	RM can view only assigned personas.	Role-based and attribute-based access control with client assignment rules.
Secrets	Use .env and never commit API keys.	Vault-managed secrets, key rotation, least privilege.
Encryption	Local demo database acceptable without real PII.	Encryption in transit and at rest, column-level protection for sensitive notes.
Logging	No raw client notes or PII in application logs.	Centralized audit logs, immutable evidence ledger, data loss prevention.
Prompt privacy	Never send unnecessary raw notes to LLM.	PII minimization, prompt redaction, provider data-processing review.
Data retention	Clear demo reset function.	Retention schedule by data class, legal hold, deletion and correction workflows.
Incident readiness	Log API failures and blocked outputs.	Breach notification workflows and operational resilience controls.

13. Suitability Guardrail Rules

The guardrail engine must run before any recommendation becomes actionable. Personalization should improve fit, not bypass investment governance.

Rule	Hard stop condition	Warning condition
Mandate	Candidate outside Defensive/Balanced/Growth mandate or worsens a known breach without reason.	Small increase in drift but still within allowed band.
Same-sector rule	Candidate not same sector where the action is framed as a same-sector swap.	Sector taxonomy differs across sources and needs RM confirmation.
CIO universe	Candidate absent from CIO-approved list or rated SELL for purchase.	Candidate rating is HOLD but not BUY.
Risk tolerance	Risk score exceeds client risk tolerance or increases by more than configured block threshold.	Risk increases within warning band.
Capacity and liquidity	Known near-term cash need conflicts with liquidity or holding period.	Liquidity need missing or outdated.
Knowledge and experience	Client lacks knowledge/experience for complex product.	Knowledge/experience unknown.

Concentration	New trade causes single-name concentration beyond hard limit.	Existing position already above warning threshold.
Sensitive data	Recommendation depends on unconfirmed sensitive inference.	Sensitive but client-provided context requires RM review.
Evidence	No evidence item supports the DNA link or market trigger.	Only one weak evidence item supports the inference.

14. Documentation and Evidence Ledger

The Evidence Ledger is the trust layer. Every generated output must be reconstructable from sources. The UI can hide complexity, but the back end must retain it.

Generated object	Minimum evidence required
Client DNA attribute	At least one RM note snippet, note date, signal type, and confidence score.
Anticipation alert	At least one trigger source plus one client-DNA link plus one portfolio link.
Swap recommendation	Current holding, candidate, CIO status, mandate check, risk check, sector check, and client-DNA alignment.
Message draft	Recommendation ID, client style tag, evidence IDs, and disclaimer that it is RM draft-only.
Meeting prep	Last interaction, open promises, portfolio changes, relevant alerts, likely questions, and source references.
Admin documentation	Client need, grounds for recommendation, suitability status, RM approval event, and final output hash.

15. Non-Functional Requirements

Category	Requirement
Performance	Cached all-clients dashboard should load in under 2 seconds. Client page should load in under 3 seconds. On-demand LLM refresh may take up to 20 seconds with progress state.
Reliability	If news or SIX API is unavailable, show cached data with timestamp and disable fresh recommendation generation.
Observability	Log latency, provider failures, cache hit rate, blocked outputs, approval events, and cost per model call.
Explainability	Every high-priority alert must include an evidence trail and the four-question action card.
Maintainability	All scoring thresholds must be stored in config, not hard-coded inside business logic.
Cost control	Precompute DNA once per client and only refresh incrementally. Cache news summaries and instrument metadata.
Testing	Unit tests for scoring formulas, guardrails, schema validation, and blocked-output cases.
Accessibility and UX handoff	API responses should include labels, confidence, status, and explanations so the front end can render calm, trust-oriented UI.

16. Recommended Build Order

Phase	Back-end build tasks	Demo value
1. Data foundation	Load CRM workbook and portfolio workbook. Normalize clients, notes, holdings, strategy, CIO list.	Shows feasibility and connects to challenge data.
2. DNA extraction	Build JSON extractor for Client DNA, Memory Card, communication style, and DNA gaps.	Creates the emotional and personalization core.
3. Signal linking	Connect holdings to news/CIO events and calculate portfolio materiality.	Creates “Why now?” and prioritization.

4. Alert engine	Generate anticipation alerts with predicted client question and four-question action card.	Makes the RM proactive and client-first.
5. Swap engine	Rank same-sector CIO-approved candidates and run guardrails.	Shows asset-level personalization inside mandate.
6. Story and action pack	Generate message, meeting prep, presentation outline, follow-up, and documentation.	Solves RM workload and makes client feel unique.
7. Audit and polish	Evidence ledger, approval state, final hashes, blocked-output rules, demo endpoints.	Wins trust, explainability, and feasibility points.

17. Visionary Extensions After MVP

Idea	Description	Why it matters
Client Reaction Predictor	Predict whether a client is likely to call, ignore, worry, or be receptive after an event.	Turns alerts into proactive relationship timing.
Relationship Temperature	Combines contact recency, unresolved promises, major life/business events, and alert urgency.	Helps the RM prioritize attention, not only AUM.
Personalization Dial	Lets the RM choose how much of the message is financial, personal, values-led, or concise.	Makes the Client Story Builder controllable and less chatbot-like.
Next-Generation Opportunity	Detects notes about children, inheritance, foundation governance, or family office transition.	Supports relationship continuity and lifetime value ethically.
DNA Drift Over Time	Shows how client preferences changed across years of notes.	Makes Client DNA dynamic rather than static.
Client Trust Replay	A one-click evidence replay showing the RM exactly why the system proposed the action.	Strengthens explainability for internal governance and pitch storytelling.

18. End-to-End Acceptance Criteria

- The all-clients dashboard ranks at least four demo clients by priority and shows why each is or is not urgent.
- Opening a client displays a Client Memory Card with values, context, communication style, previous objections, and evidence.
- At least one alert per demo scenario answers: Why now, why this client, why this action, and how to say it.
- At least one scenario proposes a same-sector, CIO-approved candidate while preserving the mandate and showing guardrail results.
- At least one scenario is blocked or downgraded because suitability, mandate, evidence, or confidence is insufficient.
- The Client Story Builder generates two visibly different message styles for the same recommendation.
- The RM approval workflow creates an immutable approval event and final output hash.
- The admin documentation includes recommendation basis, client need, evidence, suitability checks, and RM action.
- No personal social-media data source is ingested or stored in the MVP.
- The system can run a full demo without reprocessing all CRM notes on every page load.

19. Source References for Team

- SwissHacks-2026/SIX-Noumena-NTT-Data GitHub challenge README, including problem, task architecture, data, technology, and judging criteria: <https://github.com/SwissHacks-2026/SIX-Noumena-NTT-Data>
- SIX Drive / company values: Customer Focus, Collaboration, Ownership, Trust: <https://www.six-group.com/dam/download/company/publications/six-drive-en.pdf>
- FINMA Guidance 08/2024 on governance and risk management when using AI: <https://www.finma.ch/en/news/2024/12/20241218-mm-finma-am-08-24/>
- Swiss revised Federal Act on Data Protection overview, effective 1 September 2023: <https://www.kmu.admin.ch/kmu/en/home/facts-and-trends/digitization/data-protection/new-federal-act-on-data-protection-nfadp.html>
- GDPR Article 4 definitions of processing and profiling: <https://gdpr-info.eu/art-4-gdpr/>

- GDPR Article 5 principles of lawfulness, purpose limitation, data minimization, accuracy, storage limitation, integrity, confidentiality, accountability: <https://gdpr-info.eu/art-5-gdpr/>
- Swiss Federal Act on Financial Services, appropriateness, suitability, and documentation duties: <https://www.newsd.admin.ch/newsd/message/attachments/41734.pdf>
- ESMA Guidelines on certain aspects of the MiFID II suitability requirements: https://www.esma.europa.eu/sites/default/files/2023-04/ESMA35-43-3172_Guidelines_on_certain_aspects_of_the_MiFID_II_suitability_requirements.pdf